Prime-field norm relations: solver audit

November12,2026. Unsigned producer evidence [pcc:unverified].

Outcome. No novel sub-exponential ECDLP algorithm was obtained. A non-generic norm representation was examined and its tempting linear shortcut was rejected by an exact Lean counterexample. The underlying divisor/norm representation is classical and is not claimed as a new algorithm.

Online reduction. For $E/\mathbb{F}_p : y^2 = f(x)$, $p > 3$, a large odd prime subgroup of order r, public generator P and nonzero target Q, sample $k = \lceil \log_2 r \rceil + \kappa$ labelled points $R_i = [u_i]P$ after receiving the target. Reject zero, repeated x-fibres and the $\pm Q$ fibre. No reusable table or amortized precomputation is used. A signed relation $\sum_i \epsilon_i R_i = Q$ gives the logarithm $\sum_i \epsilon_i u_i \bmod r$.

Set $F = \prod_i (X - x_i)$, $G = F(X)(X - x_Q)$ and $d = k + 1$. The written geometric reduction seeks

$$A^2 - fB^2 = cG, \quad c \neq 0, \qquad A(x_Q) - y_Q B(x_Q) = 0,$$

$$\deg A \leq \lfloor d/2 \rfloor, \qquad \deg B \leq \lfloor (d-3)/2 \rfloor.$$

The simple roots and pole orders of x, y identify $\operatorname{div}(A + By) = \sum_i (\epsilon_i R_i) + (-Q) - d(O)$. The genus1 divisor/group correspondence gives both directions. Full assumptions and reasoning are in DERIVATION.md; this complete geometric reduction is not formalized in Lean.

Density is not search. For unfiltered independent labels, if X counts signs hitting fixed nonzero Q and $S = 2^k$, then

$$\mathbb{E}X = S/r, \quad \mathbb{E}X^2 = S/r + S(S-2)/r^2, \quad \operatorname{Var} X = S(r-2)/r^2.$$

Opposite signs cannot both hit Q; nonopposite rows are independent over $\mathbb{F}_r$. A fresh sample is invalid or lacks a witness with probability at most $(r-2)/2^k + (k^2+2k)/r$. This is a written derivation, not a measured solver success rate. A random sign still hits with probability $1/r$.

The missing rank-one condition

Lift coefficient products into a symmetric matrix M. Norm coefficient matching and the target branch give structured linear equations, but a true witness requires nonzero rank one. The free norm scalar absorbs the scale of a symmetric rank-one matrix; positivity over a finite field is not assumed.

For two supports normalize $B = 1$, $A = v(X - q) + y$, $Q = (q, y)$ and $U = q + x_1 + x_2$. Linear matching gives

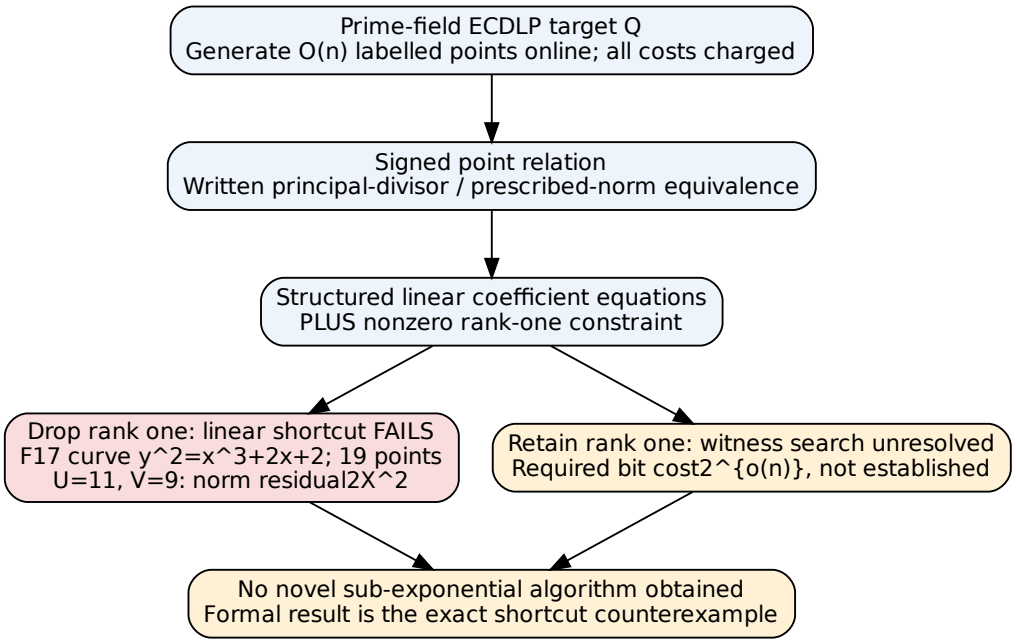
$$2yv = q^2 + a + qU - x_1x_2.$$

Under that constraint and $y^2 = q^3 + aq + b$, Lean proves

$$A^2 - f + (X - x_1)(X - x_2)(X - q) = (v^2 - U)(X - q)^2.$$

Thus the norm equation holds exactly when $v^2 = U$.

Exact prime-field counterexample. On $y^2 = x^3 + 2x + 2$ over $\mathbb{F}_{17}$, take $(5, 1), (6, 3)$ and $Q = (0, 6)$. Lean checks nonsingularity, all point equations and18 affine points plus $O = 19$. The linear solution has $U = 11, v = 9$, but $v^2 - U = 2$ and the norm residual is $2X^2 \neq 0$. The matrix $\left[\begin{smallmatrix} 11 & 9 \\ 9 & 1 \end{smallmatrix}\right]$ has determinant15. The linear equation has unique slope9, so no normalized norm witness is missed by this counterexample. It rejects the relaxation only.



Full cost and evidence boundary

Required cost. With $n = \lceil \log_2 r \rceil$ and $\log p = O(n)$, online point generation, polynomial construction, witness/sign extraction and final scalar verification have polynomial bit cost. The missing constructive norm/rank-one witness algorithm must cost $2^{o(n)}$, including failed attempts. The instance has $\Theta(n^2)$ coefficient bits: merely sub-exponential time in that larger input size is insufficient. Direct sign search costs $2^k \operatorname{poly}(n)$, still exponential. A feasible relaxed matrix or a high yes-rate decision routine does not construct the witness.

Formal scope. Mathematical source `476fe06d1`. Lean4.35.0 default build3139 jobs; actual tracked-source audit15 declarations, 0 additional trusted axioms. Generic defect/rank-condition equivalence and finite fixture facts are formal. Full divisor reduction/probability/cost arguments are written derivations. Fresh import-inclusive status is linked by the final Markdown report.

Canonical unsigned PCC records (prefix `PCC-20261112-`): ee178f defect, ef0ca9 equivalence,1b689b nonsingularity,3325a1 point count,aa5a10 points, e1609f counterexample; written derivation070fa2. Exact declaration hashes and failed/successful logs are retained. All claim types are interpretive and all standing remains unsigned.

Identity record `IDC1h8b0c962f231d89d5` binds the unconditional defect expression and formal source hash. The existing native issuer/checker accepts 32 statement-derived and32 separate checker-stream points. That checker does not execute Lean; the universal Lean identity is the algebraic proof. No random-point confidence number is used for an algorithm claim.

Sources. Galbraith, *Mathematics of Public Key Cryptography*, author-hosted version2.0,7.7.12/proof7.7.11,7.9.8–7.9.9,8.4 and10.3.6–10.3.12. PDF `https://www.math.auckland.ac.nz/~sgal018/crypto-book/main.pdf`. SHA256 `ca75ee62b96daa0d46faaa2d4dc28205683455ad3a2a48586b9b39299c25022d`. Internal prime-field/representation reviews and lead atlas informed the search. KB and ecdb unavailable; bounded search is not evidence of novelty. No priority is asserted for the norm or rank-completion formulation.

Disposition. Reject Gaussian elimination as a complete solver for this representation. A stronger constructive structured rank-one method remains unestablished. No empirical campaign, target solve, speedup, amortized setup or unrestricted impossibility theorem is reported. Existing graphs/data unchanged.